

Trevor Dayal

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Mechanical Engineering student seeking internships and co-ops for 2027 and beyond.

Education

Iowa State University of Science and Technology

Expected Graduation June 2029

Bachelor of Science in Mechanical Engineering

Experience

Formula SAE Cyclone Racing | Powertrain Simulation & Optimization Engineer September 2025 – Present

- Optimized motor design for 11 parameters using OpenMDAO.
- Krige large motor surrogate datasets to reduce optimization time.
- Implemented an AI code workflow to rapidly prototype and validate code.
- Developed lightweight full-vehicle simulator to model powertrain design.
- Prepared validation tests to confirm powertrain simulation accuracy.

FIRST Robotics (FRC) Team 4192

January 2023- April 2025

- Captained team 4192 in 2025, managed goal setting and deadlines on a 30-person team to produce a competitive robot in a 6-week period.
- Led robot design using Onshape to design a robot over a 3-week period.
- Applied lean-manufacturing techniques to reduce fabrication time by >40%.
- Reduced robot cost by over 50% through standardized part design.
- Implemented composites and 3D printed parts to reduce weight and cost.
- Manufactured parts using CNC Router, Waterjet, 3D Printers, and Laser Cutter.
- Awarded the highest possible award at 4 events.
- Nominated for FIRST Leadership Award at the 2024 Plano District Event.

Software

- **CAD:** SolidWorks Modelling, Onshape, NX Designcenter, Ansys Motor-CAD, Inventor, AutoCAD
- **Programming:** Python, Java
- **Fabrication:** Fusion 360 CAM, CNC Router, CO₂ Laser Cutter